

An Operating Condition-Based Scheme to Alternate Between Control Strategies for Improved Steady-State and Transient Behavior

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Abstract—The control strategies that use modulation techniques have great acceptance in the current control of static power converters. Indeed, they offer a wide variety of alternatives to address the steady-state characteristics required by the system. Nevertheless, its computation and modulation delay negatively affects its dynamic performance. In order to improve the time response of the system, the modulator is avoided and the strategies of direct choice of the switching state have been presented. They can provide a very fast dynamic response; however, the absence of switching state sequences makes more complex the task of establishing specific switching features for improved steady-state performance. In this work, a control scheme that selects a specific control strategy according to the operating condition of the system is proposed. This allows using modulators to satisfy the switching characteristics required in steady-state and avoids its use when a rapid dynamic response is required. In addition, a soft transition method between the control strategies is presented to prevent unwanted behaviors when these are alternated. The proposed control scheme allows simplifying the design and keeps the main advantages of each selected strategy. To validate the scheme, experimental results are obtained for a control structure that alternates between a finite set model predictive control (direct type for rapid dynamic response) and a proportional integral (PI) control with sinusoidal pulse width modulation (SPWM) (modulated type for high steady-state performance).

Index Terms—Alternating mode control, current control, finite set model predictive control (FS-MPC), proportional integral-with sinusoidal pulse width modulation (PI-SPWM), predictive control, pulse-width modulation.

I. INTRODUCTION

CONVENTIONAL applications based on a three-phase converter employ a cascaded control structure whose inner loop is a current controller. Consequently, the current

control design is an important topic in power electronics since largely defines the system performance. Certainly, the converter is a hybrid system; therefore, the performance criteria of the control strategy are not only referred to its speed and adequate current regulation. It must also consider the efficiency of the converter and the load, established mainly by its steady-state performance. There are recognized strategies that provide a solution—of simple concept and implementation—to some of these performance criteria. On the one hand, the strategies that use modulation techniques have been highlighted to meet the steady-state performance criteria [1]–[8] and on the other hand, there are strategies that lack a modulator and are highlighted by its rapid reference tracking [9]. The challenge is to satisfy both requirements in a single scheme. However, such schemes become more complex and computationally expensive. In this context, the availability of faster processing hardware at lower cost has facilitated the implementation and it has motivated the research on new increasingly sophisticated control strategies [10]–[13]. Nevertheless, it is still a challenge to design a single control strategy that meets both criteria, steady-state performance and dynamic response, as independently performed by strategies with and without modulator, respectively.

This work addresses the problem from the restrictions and the main advantages presented by the strategies with and without modulator. It is argued that a strategy has a better performance and a simpler design when it is not attempted to pursue simultaneously the criteria of steady-state and fast dynamic. Instead, it is recommended to distinguish between two system-operating conditions: when it is required that the strategy presents a rapid dynamic response and when it is not. Thus, it is proposed to operate with a different control strategy in each operating condition, each one dedicated exclusively to meet the requirements of the system condition in which it operates. The main challenges now lie on establishing criteria to identify each operating condition and a method to avoid undesirable behaviors of the control strategies during the transition.

To validate the proposed control scheme, transitions between a finite set model predictive control (FS-MPC) and a linear modulator-based control with proportional integral with sinusoidal pulse width modulation (PI-SPWM) (FS-MPC) are performed, as whether a rapid dynamic response or not is, respectively, required. Experimental results are obtained in a three-phase voltage source converter [2L-voltage source inverter (VSI)] with an RL load. However, the proposed procedure can be extended to any topology and control strategies.

Manuscript received July 29, 2014; revised June 27, 2015; accepted September 08, 2015. Date of publication September 23, 2015; date of current version December 02, 2015. This work was supported in part by the Chilean Government under Project CONICYT/FONDECYT 1120683 and Project CONICYT/FONDECYT 1140594, and in part by the Solar Energy Research Center (SERC) Chile under Project CONICYT/FONDAP/15110019. Paper no. TII-15-0284.

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Digital Object Identifier 10.1109/TII.2015.2481791

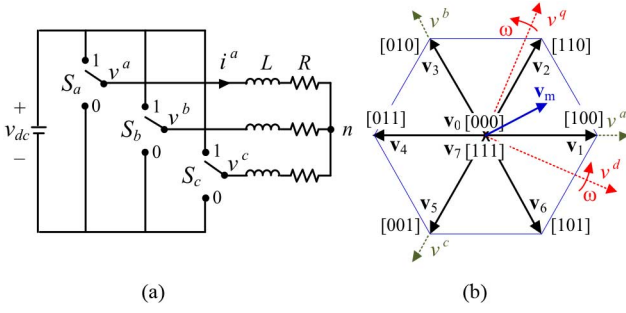


Fig. 1. Two-level voltage source inverter (2L-VSI). (a) Topology. (b) Voltage vectors v_n and their respective switching states $s_n \in S_a, S_b, S_c$.

II. BACKGROUND

A. Performance Criteria

The study case of this work is the inner loop of current control that receives the references from an outer control loop. Performance criteria of the inner control loop consist in obtaining a rapid dynamic response and specific switching characteristics in steady-state. The required switching characteristics are aimed to improve the efficiency of the converter and the load in the context of a specific application. Among the typical examples of these requirements are fixing or reducing the switching frequency, to avoid the presence of certain harmonics in the current, and minimizing the harmonic distortion.

B. System Model

The typical representation of the two-level voltage source inverter (2L-VSI) is presented in Fig. 1(a). Each phase can assume two states, represented by the integers $S_a, S_b, S_c \in \{0, 1\}$. The three-phase amounts of voltage v , current i , and switching state s can be represented as space vectors. By expressing them in abc axes $v^{abc} = [v^a \ v^b \ v^c]^T$, $i^{abc} = [i^a \ i^b \ i^c]^T$, and $s^{abc} = [S_a \ S_b \ S_c]^T$ are obtained. The converter supports eight switching states $s_0 - s_7$. This allows achieving seven different voltage values: six active voltages $v_1 - v_6$ and two null voltages v_0 and v_7 . The voltage vectors and their respective switching states are presented in Fig. 1(b).

The system model is obtained according to the control strategy.

1) *Converter Model for Direct Control Strategies:* These strategies have as control action the switching state s of the converter. Then, its model is

$$v_n(t) = s_n(t) \cdot v_{dc}(t) \quad (1)$$

where v_{dc} is the dc voltage and n indicates the chosen switching state and its respective voltage vector.

2) *Converter Model for Modulated Control Strategies:* These strategies have as control action the mean voltage applied into the load v_m . A modulation technique implements v_m by means of predefined combinations of switching states. The input of the modulator is the modulating signal m and its relation with v_m is given by

$$v_m(t) = G_m \cdot m(t) \cdot v_{dc}(t) \quad (2)$$

where G_m is the gain of the modulation technique.

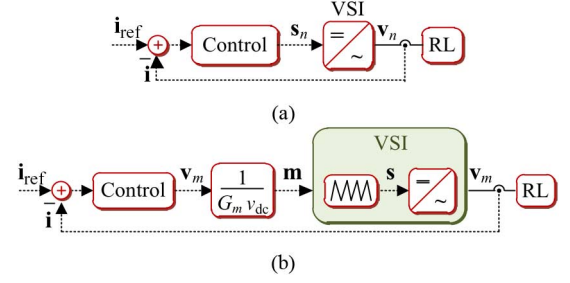


Fig. 2. Control strategies. (a) Direct. (b) Modulated.

3) *System Model:* In this work, the control strategies are assessed by using an RL load. Then,

$$L \frac{d}{dt} i(t) = v(t) - Ri(t) - j\omega Li(t) \quad (3)$$

where R – L are the resistive and inductive parameters of the load and ω is the angular speed of the reference framework.

The system model is obtained from (3) by replacing the voltage v by either v_n of (1) or v_m of (2), accordingly. In Fig. 2, two block diagrams of the described control strategies are presented where the dq axes representation is preferred. Then, given a vector expressed in the abc axes $x^{abc} = [x^a \ x^b \ x^c]^T$, its equivalent in dq axes $x^{dq} = [x^d \ x^q]^T$ is

$$x^{dq}(t) = T_{abc2dq}(t) \cdot x^{abc}(t) \quad (4)$$

with

$$T_{abc2dq}(t) = \frac{2}{3} \begin{bmatrix} 1 & e^{j(2\pi/3)} & e^{j(4\pi/3)} \end{bmatrix} e^{-j\omega t}. \quad (5)$$

III. ELECTION OF THE CONTROL STRATEGY

In this section, both direct and modulated strategies are revised from the perspective of the election of switching states, highlighting some of the advantages and constraints imposed by the control action.

A. Direct Strategies

Using s_n as a control action, involves to keep v_n applied—and thus maintaining the same trajectory of the current—during the whole sampling time T_s . This restricts the strategy to choose one of the seven available trajectories in each T_s , as noted in Fig. 3(a). In order to achieve a required variation in the current Δi_{ref} , only the application of combinations of v_n can be sought. To achieve more precise currents variations Δi , it becomes convenient to implement these strategies with smaller T_s . Thus, the total application time of each v_n can be chosen with greater precision, as presented in Fig. 3(a). However, the resulting precision is not comparable with that obtained with modulators, because the application time of v_n is always a multiple of T_s .

The selection of s_n must consider performance criteria related to switching and speed. A common feature of the performance criteria related to switching characteristics is the fact that they assess a specific kind of relationship existing between

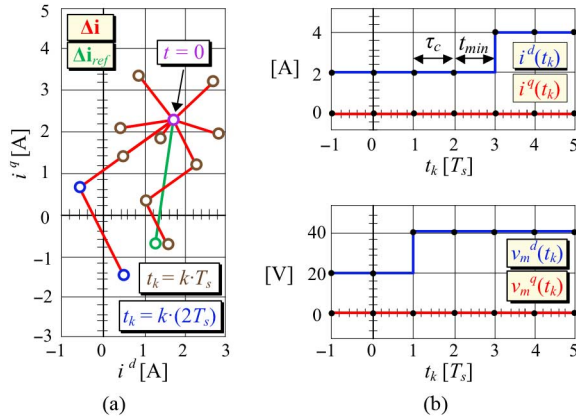


Fig. 3. Restrictions according to the control action used. (a) Resulting trajectories according to the v_m available at direct and precision strategies, according to chosen T_s . (b) Effect of delay by compensation of the τ_c calculation and minimum response time t_{min} , in the fundamental i current in modulated strategies.

the switching states applied during a specific time window. Then, to meet these criteria, the selected s_n must be somehow related to the previously chosen s_n . Contrary, for a fast dynamic response, the s_n that minimizes the error in the shortest possible time must be chosen without considering if it is related to other s_n that have already been applied. If both requirements are pursued simultaneously, they confront each other, because the chosen s_n can meet the switching criterion, but not minimizing the error or vice versa.

Regarding the performance, these strategies are highlighted by their dynamic response when they choose the s_n that minimizes the error, neglecting the switching characteristics [9]. It is possible to improve the switching characteristics; however, the confrontation between the objectives requires either to weight one over the other one or to balance them. Predictive strategies that assign a weighting factor to each objective inside the same cost function have been suggested. Thus, it has been proposed to implement predefined sequences of s_n [14]–[18], to use a filter to involve s_n applied prior to the election [19], to punish the number of switching involved [20], and associated common mode voltage [21], [22] among others [10]–[13]. These proposals increase the complexity of the algorithms without obtaining a steady-state performance (switching characteristics and zero steady-state error) similar to those of the modulated strategies.

To avoid the conflict of objectives, it has been proposed to discriminate the condition of the system by using hysteresis bands [10], [12]. It is sought to pursue one criterion at a time according to the operating condition, allowing improving the performance of each condition of the system. Moreover, by not imposing predetermined sequences of s_n , a spread spectrum is obtained. Thus, these strategies aim to reduce the switching frequency.

B. Modulated Strategies

The modulator allows decoupling the system dynamics that addresses the control strategy of the dynamics associated with the switching characteristics. The modulator samples v_m and implements it using a sequence of different s_n during a time

window T_m . This implies that the strategy must choose a single v_m during all T_m . Then, if the choice is made at regular times, the sampling time of the strategy T_s matches with T_m . This takes place even in the particular case of SPWM, whose common practice is to implement it with regular samplings [23]–[25]—at the beginning of each modulation cycle or subcycle [28]—in order to achieve proper sampling of the fundamental current component.

The choice of v_m must consider only the dynamic requirements of the system. The minimum response time t_{min} of the strategy is T_s because the control action response v_m is obtained in the following sampling instant. In addition, the implementation of these strategies should consider the computation delay τ_c , whose compensation adds a T_s delay to the response time. These characteristics constrain the minimum response time of modulated strategies, as depicted in Fig. 3(b), being more critical when predefined sequences of s_n are implemented in large T_m . This impacts negatively the dynamic performance and presents a greater challenge to the design of the control strategy [23]–[27].

The required T_m for implementing v_m depends on the modulation technique used. When it is intended to set the switching frequency to f_{sw} —like in the case of SPWM or SVM— T_m can have a duration of $1/f_{sw}$ or even of $1/(2 \cdot f_{sw})$, depending on whether the pre-established frequency is related to a cycle or semicycle of modulation. When the optimization of the sequences of s_n is sought, frequencies obtained offline whose T_m has a duration of a quarter of wave are usually implemented [4]–[8]. Regarding performance, these strategies are highlighted by their stationary performance where a wide variety of modulation techniques that meet different requirements in the switching characteristics are available. There are a number of studies that compare their dynamic response with that of direct strategies [29]–[31]. Nevertheless, the smaller the T_m —such as SPWM with high f_{sw} —the lower is the minimum response time, enabling to achieve rapid dynamic responses [27].

IV. ALTERNATING CONTROL MODE

There are two specific operating conditions of the system: 1) *dynamic condition*; and 2) *stationary condition*. The dynamic condition occurs when a rapid dynamic response is needed by the system. When the system is not in dynamic condition, the stationary condition is assumed, where the strategy can be entirely devoted to establish the best switching characteristics. By assuming that the dynamic condition is transitory, it is possible to neglect the expected switching characteristics during such condition without affecting considerably the overall efficiency criteria of the system.

Indeed, this work proposes a new method to meet the performance requirements by dedicating a specific control mode (CM) to each operating condition and alternating between the modes conveniently. Thus, the stationary control mode (SCM) and dynamic control mode (DCM) are defined. These are devoted only to meet steady or dynamic criteria, respectively.

The main control algorithm must be capable to conveniently select only one CM to operation. In Fig. 4, a CM algorithm flowchart of the main control scheme is presented. Due to practical reasons, a supervising stage called alternating control

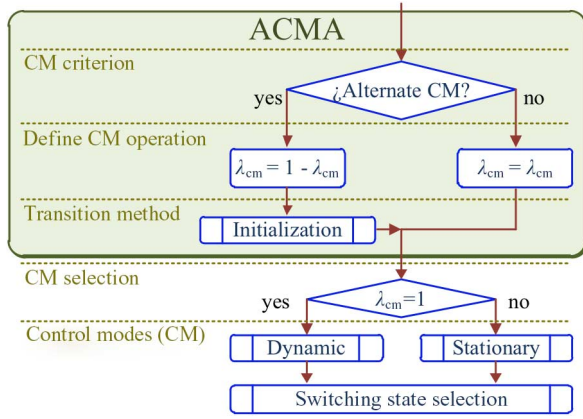


Fig. 4. Alternating CM algorithm flowchart.

mode algorithm (ACMA) is defined, which selects the CM by using a Boolean variable λ_{cm} . Unlike alternation methods based on hysteresis bands, the proposed scheme allows both direct and modulated strategies in any of the CMs, each one implemented with its own sampling time.

The design of the main control scheme requires the following.

- 1) Define criteria or rules to alternate between CMs.
- 2) Choose the control strategies that will operate in each CM.
- 3) Define transition methods between CMs.

These issues are highlighted in Fig. 4 and described in more detail in the following sections.

It should be noted that if the method to alternate between CMs ensures a soft transition and an independent operation of each CM, the stability of the whole control scheme depends on the individual stability characteristics of each strategy. Even if the method incorrectly selects a CM, the performance will be governed by the characteristics of the associated strategy.

A. Control Strategies

The proposed scheme allows that each CM operates independently. Then, if the transition between CMs and the alternating CM criteria is adequate, the performance of the scheme will be determined by the individual performance of each CM. Hence, the choice of the control strategies that operate in each CM must be in accordance to the requirements of the system.

In order to demonstrate that it is possible to use both direct and modulated strategies, in this work a PI linear control with SPWM in SCM and FS-MPC in DCM are used. The reason of the choice is the remarkable performance presented by each strategy in stationary and dynamic condition, respectively. Both strategies are briefly described as follows.

Linear Control With SPWM:

To design the linear control, an auxiliary variable is defined

$$\mathbf{u}^{dq}(t) = \mathbf{v}_m^{dq}(t) - j\omega L \mathbf{i}^{dq}(t) \quad (6)$$

and by replacing in (3) the linear relation

$$\mathbf{u}^{dq}(t) = R \mathbf{i}^{dq}(t) + L \frac{d}{dt} \mathbf{i}^{dq}(t) \quad (7)$$

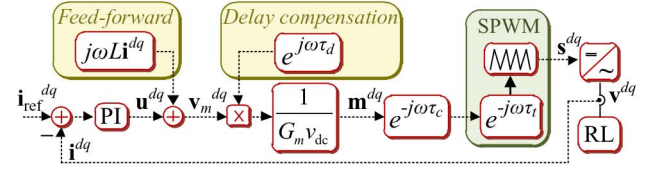


Fig. 5. PI-SPWM signal flow graph with feed-forward and delay compensation.

is obtained allowing to design the current control loop with a PI controller, whose equation of differences is

$$\mathbf{u}^{dq}(k) = \mathbf{u}^{dq}(k-1) + C_1 \cdot e^{dq}(k) + C_2 \cdot e^{dq}(k-1) \quad (8)$$

where C_1 and C_2 are constants defined by the controller tuning. The control action $\mathbf{v}_m$ is obtained from the output of the controller $\mathbf{u}^{dq}$ using (6) as a feedforward signal.

It is a common practice that the implementation of the strategy is carried out with a regular sampling to acquire only the fundamental component of $\mathbf{i}$ [25]. In addition, the compensation of the computation delay τ_c and modulation delay τ_t found in carrier-based modulators must be taken into account. Indeed, the modulation delay is equivalent to $\tau_t = 0.5 \cdot T_m$ and by being implemented with regular sampling, the total delay associated with the control strategy is $\tau_d = \tau_c + \tau_t = T_s + 0.5 \cdot T_s = 1.5 T_s$ [24], [25]. The block diagram of the proposed strategy with compensation is depicted in Fig. 5.

The delay becomes more significant when operating at low f_{sw} , presenting a major challenge in the design if obtaining a rapid dynamic response is sought [23]. Because this CM does not require such dynamic response, the tuning presents no further considerations. Indeed, no method is needed to deal with saturation, such as antiwindup [32].

1) *Finite Set Model Predictive Control*: FS-MPC is a direct strategy, characterized by choosing the switching state by solving an optimization problem, taking advantage from future information of the system. For the current control of the 2L-VSI that feed an RL load, the future information can be obtained by means of the following predictive model [9]:

$$\mathbf{i}_n^{dq}(k+1) = \mathbf{i}^{dq}(k) \left[1 - \frac{RT_s}{L} - T_s \mathbf{W} \right] + \frac{T_s}{L} \mathbf{s}_n^{dq}(k) v_{dc}(k) \quad (9)$$

using (1) and (3) and the Euler approach.

The optimization problem is presented as a minimization of the cost function that represents the control objective of the strategy. This is solved by evaluating each prediction in the cost function and selecting the $\mathbf{s}_n$ that has reached the minimum value. In order to achieve a rapid dynamic response, it is possible to define the cost function as [21]

$$g = |\mathbf{i}_{ref}^d - \mathbf{i}^d| + |\mathbf{i}_{ref}^q - \mathbf{i}^q|. \quad (10)$$

The procedure of the conventional FS-MPC consists in determining $\mathbf{i}_n(k+1)$ for every $\mathbf{s}_n$ using (9) and then, evaluating each one of them in (10). It is chosen as the optimal switching state $\mathbf{s}_{opt}$, the one that minimizes the cost function (10) and finally it is applied to the converter. The calculation delay

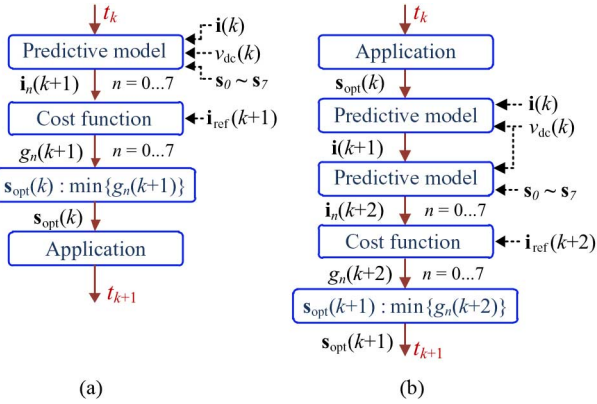


Fig. 6. FS-MPC algorithm flowchart for current control in a 2L-VSI. (a) Conventional. (b) With computation delay compensation.

also presents deterioration in the performance of this strategy and the procedure is modified to compensate it [33]. Both procedures are presented in Fig. 6.

B. Alternating CM Criteria

In order to identify different operating conditions of the system, the control schemes based on hysteresis band use an error threshold as criterion. Thus, if the ACMA is used, the hysteresis criterion can be defined as

$$\lambda_{cm} = \begin{cases} 1, & \text{if } |i_{ref}(k) - i(k)| > \delta_h \\ 0, & \text{if } |i_{ref}(k) - i(k)| \leq \delta_h \end{cases} \quad (11)$$

where δ_h is the maximum permitted error in stationary condition.

It is necessary to consider that if the selection of the switching states is made by a modulator, the magnitude of the current error is affected by the operating point $\{i, v_{dc}\}$ of the system and the characteristics of the selected modulation technique. Then, if the maximum current error is not bounded in stationary condition, the criteria established in (11) cannot be used since it may produce undesired alternations between strategies as a result. Thus, the hysteresis criterion (11) is effective only if the operating strategy of SCM includes the threshold value in its design and aims to maintain the current bounded as the criteria stated. Considering the above, a different alternating CM criterion must be proposed to avoid the supervision of the operation or performance of the implemented strategy. Thereby, unwanted alternation between CMs driven by an incorrect interpretation of the normal operation of the strategies can be avoided. Also, it will become unnecessary to include the alternation criteria in the strategy design.

Different criteria are proposed according to each CM, as indicated below.

- 1) *Alternating to DCM*: Being in a stationary condition, there are two situations by which an alternation to DCM could be required; a change in the reference value or a disturbance in the system. Since the current control is designed to operate as a cascaded loop, any disturbance will be corrected by the outer control loop by changing the reference value. Therefore, unless there is a significant

change in the reference, the SCM must be maintained in operation. Then, the proposed criterion to alternating to DCM is

$$|\Delta i_{ref}^{dq}(k)| \geq \delta_i \quad (12)$$

where δ_i indicates the maximum change in the reference value permitted in stationary condition and $\Delta i(k) = i(k) - i(k-1)$.

- 2) *Alternating to SCM*: To alternate to DCM, the reference value must have changed (12) indicating a different desired operating point. Then, to consider returning to SCM again, the desired operating point must be reached. It means that the current error must be minimized and also there must be no significant changes in the reference, i.e., (12) must no longer be fulfilled. Subsequently, the proposed criteria to alternate to SCM are

$$|\Delta i_{ref}^{dq}(k)| < \delta_i \quad (13)$$

$$\min \left\{ |i_{ref}^{dq}(k) - i^{dq}(k)| \right\}. \quad (14)$$

Once the significant changes in the reference are no longer produced, i.e., (13) is fulfilled, (14) needs to be satisfied only once to alternate to SCM. Expression (14) could be implemented comparing the error magnitude with a certain threshold, as occurs in (11); however, the election of the threshold δ_h is not trivial. Therefore, it is proposed to implement (14), detecting the moment where the current intersects the reference value. In practice, this can be performed by detecting the moment where the error of the current changes its sign. Hence, the *sign* function is used

$$\text{sign}(x) = \begin{cases} 1, & \text{if } x \geq 0 \\ -1, & \text{if } x < 0. \end{cases} \quad (15)$$

The procedure consists in storing the error when (13) is fulfilled by

$$\rho = \text{sign} \{ i_{ref}(k) - i(k) \} \quad (16)$$

and in the following sampling instants, it must be evaluated only whether a change of error sign occurs

$$\rho \cdot (i_{ref}(k) - i(k)) \leq 0. \quad (17)$$

It should be noted that (16) and (17) are applied for the d and q components of the current vectors i^{dq} and i_{ref}^{dq} .

C. Transition Between CMs

In order to compute the actual control action, some strategies require stored information of the evolution of internal variables. When an alternating CM occurs, the selected strategy was inactive; thus, it has not stored such information and may exhibit an initial involuntary behavior. To avoid this malfunctioning, the internal variables of the selected controller should be rewritten with adequate information.

Regarding the chosen strategies, the FS-MPC only requires current and future information for its operation. Therefore, it

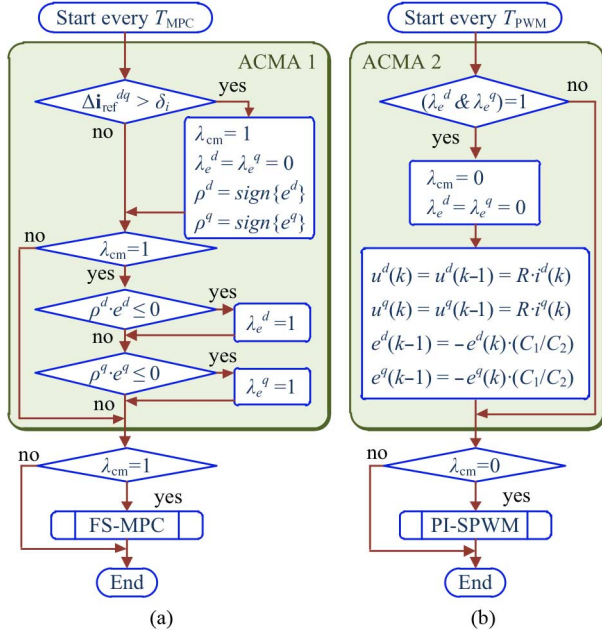


Fig. 7. Flowchart of the algorithm of the proposed control sequence. (a) FS-MPC interruption. (b) PI-SPWM interruption.

is not necessary to provide additional information. Contrary, the case of the linear controller PI (8) requires $u^{dq}(k-1)$ and $e^{dq}(k-1)$ to determine $u^{dq}(k)$.

Since the PI strategy operates as SCM, the information that must be provided has to describe a stationary evolution of the system at the operating point where the alternating CM occurs. Thus, it is sought that the initial behavior of the strategy maintains such operating point. In order to achieve this, from (7) it is possible to obtain the output of the controller in steady-state

$$u^{dq}(0) = R i^{dq}(0). \quad (18)$$

Then, to describe a stationary evolution, it must be hold that

$$u^{dq}(k) = u^{dq}(k-1) = u^{dq}(0). \quad (19)$$

By using (19), $e^{dq}(k-1)$ can be obtained from (8)

$$e^{dq}(k-1) = -\frac{C_1}{C_2} \cdot e^{dq}(k) \quad (20)$$

because the current error is unknown.

Summarizing, every time an alternating CM occurs and the SCM is selected, $u^{dq}(k-1)$ and $e^{dq}(k-1)$ should be initialized to avoid an undesired initial behavior of the strategy.

D. Alternating CM Algorithm

ACMA is responsible for assigning a value to λ_{cm} and thus, conveniently alternates the CM, as stated by Fig. 4. To allow a different sample time operation of each selected strategy, the algorithm is implemented in two different interruptions, as illustrated in Fig. 7. Also, ACMA is divided into two stages, ACMA 1 and ACMA 2, to ensure that the alternation occurs synchronized with the beginning of the corresponding strategy. The aim of ACMA 1 is to detect dynamic condition and

TABLE I
EXPERIMENTAL SETUP PARAMETERS

Variables	Description	Values
P^{nom}	2L-VSI nominal power	20 (kW)
V_{dc}	DC-link voltage	100 (V)
P_L^{nom}	Load nominal power	1 (kW)
L	Load inductance	15 (mH)
R	Load resistance	10 (Ω)
f_o	Output frequency	50 (Hz)
T_{PWM}	Sampling period of PWM interruption	1 (ms)
f_{sw}	Commutation frequency	1 (kHz)
T_{MPC}	Sampling period of MPC interruption	100 (μ s)
f_{sw}	Commutation frequency	1 (kHz)
k_p	Discrete-time proportional gain	4.415
k_i	Discrete-time integral gain	1.887
δ_i	Threshold of current reference	0.5

is described in Section IV-B1). The aim of ACMA 2 is to detect stationary condition and is described in Section IV-B2), adding the initialization stage illustrated in Fig. 4 and described in Section IV-C. This initialization stage ensures the adequate transition between CMs. As described in the Fig. 7, the Boolean variables λ_e^d and λ_e^q are conveniently defined to alternate the mode after both error components have minimized the error in dynamic condition, as shown in (16) and (17). In addition, it should be noted that only the ACMA and one of the strategies are computed at all times, which minimizes the computational burden.

It should be noted that the alternating method proposed has as main criterion the evaluation of the dynamics of the reference. Consequently, only one parameter δ_i must be defined and its configuration does not depend on the model of the system neither of the design of the chosen strategies. The magnitude of δ_i indicates the minimum magnitude step change of the reference that must occur to begin the operation of a rapid dynamic response strategy.

V. VALIDATION

In this section, the operation of the proposed control scheme is validated by describing the behavior of the internal variables and their performance.

A. Experimental Prototype

An experimental prototype of the 2L-VSI was built-up with an *RL* load as illustrated in Fig. 1. The associated parameters are illustrated in Table I. The control scheme is implemented in a dSPACE 1103 platform, connected to a PC for programming and data acquisition. The FS-MPC is implemented with a sampling time of $T_{MPC} = 100$ (μ s), whereas the PI-SPWM with $T_{PWM} = 1$ (ms), $f_{sw} = 1$ (kHz), and $\omega_{BW} = 100$ (Hz). The control algorithm of the PI is run by the main processor of the dSPACE 1103, whereas the SPWM is implemented in a TMS320F240 slave processor provided by the platform to obtain a better resolution of the width of the pulses. The parameter δ_i is initially fixed in 0.5. However, this decision will be discussed below.

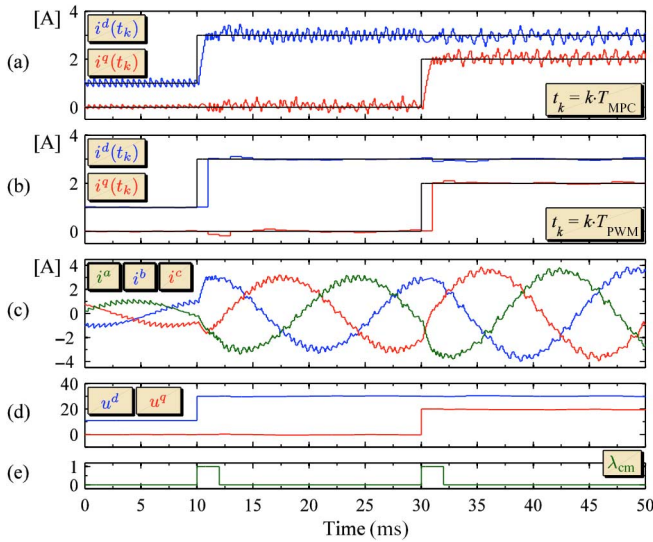


Fig. 8. Performance of the proposed control scheme. (a) i^{dq} sampled at T_{MPC} . (b) i^{dq} sampled at T_{PWM} . (c) Three-phase current i^{abc} . (d) Output of the PI controller u^{dq} . (e) Flag of the switching CM λ_{cm} .

B. Performance

In order to assess the performance of the proposed scheme, step changes of the current i^d and i^q are performed, as indicated in Fig. 8. The experimental currents sampled by each strategy at T_{MPC} or T_{PWM} are shown. The variable λ_{cm} indicates with a unitary value if the FS-MPC algorithm is operating; otherwise, the PI-SPWM algorithm operates. The detection of the reference change is implemented by ACMA 1 at every T_{MPC} , which allows alternating to DCM rapidly and avoid the characteristic delay of the PI-SPWM.

Since FS-MPC is operating in dynamic condition, a fast and decoupled error minimization is achieved [9], theoretically illustrated in Fig. 3. Although in transient, the PI-SPWM control is not in operation, its internal variables are initialized according to the new reference value in order to be ready for an adequate transition to SCM. This can be evidenced in the value taken by the output controller u^{dq} in transient, which coincides with the value that subsequently calculates the PI controller for the steady-state. The current sampled at T_{PWM} evidences the regular sampling, because only its fundamental value is sampled [23]–[25].

In particular, it must be noted that a deadbeat type dynamic response is achieved, because the transient occurs during $T_{PWM} = 1$ (ms) approximately. However, it is highlighted that the scheme allows achieving responses faster than the deadbeat control by avoiding its characteristic delay.

C. Operation Principle

A step change in i^d is shown in Fig. 9 to illustrate how the alternating CM operates. By evaluating $|\Delta i_{ref}^{dq}|$ it is possible to detect instantaneously the change of i_{ref}^{dq} . In particular, it is shown that when $|\Delta i_{ref}^{dq}|$ exceeds δ_i , the value λ_{cm} changes to 1 and consequently, the CM is alternated. Indeed, in that instant, the sign of the error in each dq component is stored (16) and later, only the error change is assessed (17), as indicated in the ACMA.

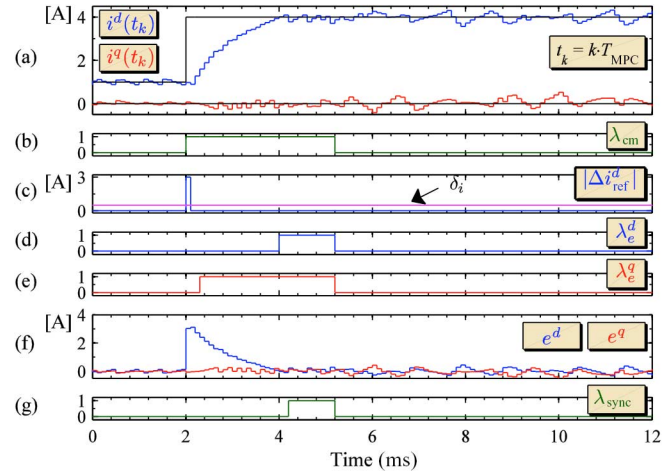


Fig. 9. Control variables of the proposed scheme. (a) i^{dq} sampled at T_{MPC} . (b) Flag of the switching CM λ_{cm} . (c) Absolute value of the variation in the reference i^d . (d) Flag indicating change of the sign in e^d . (e) Flag indicating change of the sign in e^q . (f) Current error e^{dq} . (g) Flag λ_{sync} indicating the delay compensation made to switch to SCM.

In addition, the error of the current sampled at T_{MPC} is shown in the Fig. 9 and it is possible to evidence that when the sign of the error changes, either λ_e^d or λ_e^q takes immediately a unitary value, accordingly. When both components of λ_e take a unitary value, it is deduced that the value of i^{dq} has reached the value of i_{ref}^{dq} . Despite being in steady-state, the DCM must continue operating until the following sampling instant of the SCM strategy occurs, enabling a synchronous alternating CM. Thus, the initial control action of the SCM can be applied to the whole following T_{PWM} sampling time.

In this particular platform, there is an additional transport delay between the assignment of a new value of the modulator and its effective implementation. When PI-SPWM is in normal operation, this transport delay has no effect because it occurs during the implementation of the computation delay τ_c . Nevertheless, if an alternating CM occurs and PI-SPWM is selected, the delay prevents the initial calculated modulator to be implemented instantly and a modulator of inadequate value is applied in the meantime. Therefore, an additional T_{PWM} time is awaited—indicated by the variable λ_{sync} —before alternating to SCM. Thus, it is ensured that the calculated initial modulator is implemented when alternating the CM. It should be noted that this additional consideration is not due to the algorithm. It is only performed by the characteristics of the dSpace platform used. Moreover, its only consequence is that the FS-MPC algorithm operates an additional T_{PWM} time in steady-state.

D. Evaluation of the Alternating CM

As the proposed scheme is based on the alternation of CMs, it is necessary to illustrate its behavior in case of an unnecessary alternation of the scheme. This might happen in steady-state if for any reason the reference change $|\Delta i_{ref}|$ exceeds for a brief instant the threshold value δ_i , as shown in Fig. 10. It is observed that in the same instant in which $|\Delta i_{ref}|$ changes its value, the CMs are alternated and the internal variables of the PI controller are initialized, as exhibited in the change of value of u^{dq} . However, as the reference returns quickly to its initial value, the

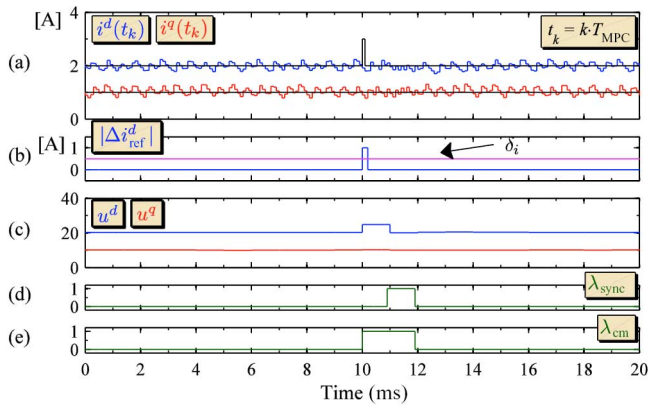


Fig. 10. Evaluation of unnecessary switching CM detection. (a) i^{dq} sampled at T_{MPC} . (b) Absolute value of the variation in the reference i^d . (c) Output of the PI controller u^{dq} . (d) Flag λ_{sync} indicating the delay compensation made to switch to SCM. (e) Flag of the switching CM λ_{cm} .

FS-MPC fails to react and maintains the operating point until the scheme allows the alternating to PI-SPWM again. Because the reference is provided by an external loop, a pulse of such magnitude should not be expected. However, it is highlighted that this has the only consequence of an unnecessary operation of FS-MPC during a brief period. This is achieved mostly due to the proposed transition method of strategies, which allows an initial behavior of PI-SPWM without disturbances.

E. Election of δ_i

In Figs. 9 and 10, the relation between the value of $|\Delta i_{ref}|$ and the threshold δ_i is shown. It is found that the results of the Figs. 9 and 10 could have been the same by using $0 < \delta_i < 3$ and $0 < \delta_i < 1$, respectively. Contrarily, if $\delta_i = 4$ would have been used, the dynamic condition could have not been detected in neither case, and the PI-SPWM could have operated all the time. The criterion of choosing $\delta_i = 0.5$ considers that it is not necessary to alternate the strategies—sacrificing the switching characteristics—when the magnitude of the reference change is comparable to the magnitude of the ripple of current in steady-state. Hence, the value of δ_i used only allows to alternate to FS-MPC when the reference change is greater than 0.5.

VI. CONCLUSION

This work proposes a new control scheme based on the alternation of strategies for current control. Therefore, new criteria to detect different conditions of the system, a method to perform soft transitions between strategies, and an algorithm to implement them are proposed. The need to alternate between strategies is argued by describing both advantages and disadvantages of modulated and direct strategies. As a result, a control algorithm is presented that requires setting a single parameter δ_i , whose value does not depend on the model of the system. Furthermore, the design of the selected strategies is simplified, because they only have to meet either stationary or dynamic requirements. Particularly, just the PI strategy is tuned, whose dynamic response is not relevant to the design.

The control scheme is validated in an experimental prototype of the 2L-VSI with an RL load. It is highlighted that the scheme detects the different operating conditions of the system at a time rate given by the faster sampling time. This allows to conveniently alternating between strategies, achieving a soft transition among them. The rapid alternation allows avoiding the inherent time delay of the modulation techniques. In particular, the scheme by alternating between FS-MPC (featuring a rapid dynamic response) and PI-SPWM (featuring proper switching characteristics) is validated. It must be noted that the approach is presented in such way that can be used with other control strategies in both dynamic and steady mode.

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